MILITARY SPECIFICATION

MINE ANTI-PERSONNEL, M86, SPECIAL TEST MINE LOADING AND ASSEMBLY OF

This Specification is approved for use within the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the loading, assembling and packing of the Special Test Mine (STM), for the Mine, Anti-personnel M86 assembly of the munition contained therein.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification For
MIL-3-70731	- Battery, Primary, Reserve Ammonia, 8 Milliampere Hour Capacity for Mine Anti-Personnel M86
MIL-M-70780	- Mine, Anti-Personnel, M86 Arming Assembly For
MIL-M-70777	- Munition for Mine, Anti-Personnel, M8 'Transfer Lead Assembly
MIL-M-48245	- Molding Compound, Housing and Plug
MIL-B-70776	- Bandoleer, M86 Mine

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to. Commander, U.S Army Armament, Munitions and Chemical Command, Attn AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1345
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202610-23T03:59Z
Check the source to verify that this is the current version before use.

MIL-M-70778 - Munition for Mine, Anti-Personnel, M86 Acceptor Lead Assembly For
MIL-M-70697 - Mine, Anti-Personnel, M86 Molded Assembly For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-286 - Propellants, Solid: Sampling, Examination and Testing

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of solicitation.

DRAWINGS (See 6.7)

U.S. ARMY ARMAMENT, RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

12579595 - Arming Assembly, PDM
9366661 - Molded Assembly, PDM
9366705 - Special Test Mine, (STM) for Pursuit Deterrent Munition
9366710 - Packing and Marking for Box, Wirebound for Mine APERS (PDM) , M86 or Classroom Model
12910122 - Special Test Mine with Bandoleer for M86 APERS Mine

(Copies of specifications, standards handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents which is current on the date of the solicitation.

AMERICAN SOCIETY OF TESTING AND MATERIALS (ASTM)

ASTM-D-695 - Compressive Properties of Rigid Plastics.

(Application for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 14103.

Source: https://assist.dla.mil -- Downloaded: 202610-23T03:59Z
Check the source to verify that this is the current version before use.

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification, however shall supersede applicable laws and regulation unless specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies.

3.2.1 Mine. The Special Test Mine (STM) shall comply with all requirements specified on Drawing (Dwg. 9366705 and 9366661) all associated drawings and with all requirements specified in applicable specifications.

3.2.2 Munition. The inert munition and components shall comply with all requirements specified on Dwg. 9298449, all associated drawings and with all requirements specified in applicable specifications including MIL-B-70731, MIL-M-70780, MIL-M-70777, MIL-B-70776, MIL-M-70778 and MIL-M-60697.

3.3 Functioning.

3.3.1 Arming. There shall be no arming of the S&A, firing of the piston actuator or the gas generator prior to 25 seconds after depressing the battery ball.

3.3.2 Safe dud. The mine, if not functioned by any external or internal mode, s'nail not function after the maximum self destruct time has elapsed.

3.3.3 Hazardous dud. All safe duds when tested as specified in 4.5.5.1, shall not be capable of functioning after the maximum self destruct time has elapsed.

3.3.4 Anti-disturbance functioning. The mine shall function in the anti-disturbance mode as the mine is disturbed by two (2) different movements to two (2) different sides.

3.3.5 Breakwire functioning. The mine shall function in the breakwire mode when any deployed tripline is pulled with a minimum force of one hundred and forty grams (140 g) to max force of six hundred grams (600 g).

3.3.6 Self-destruct functioning. The mine shall be capable of self-destructing itself within the self-destruct minus 20% of time limit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:59Z
Check the source to verify that this is the current version before use.

3.3.7 Early functioning. The mine shall not self-destruct itself by any internal mode within 80% of the self-destruct time.

3.3.8 Tripline deployment. The mine after ground impact and with unobstructed field, shall deploy a minimum of three (3) triplines with at least one (1) deployed tripline attaining-a minimum of deployed length of thread of ten (10) feet and two (2) others providing additional length to total sixteen (16) feet for three (3) triplines.

3.4 Documentation of plan for safety in handling of munitions. Prior to loading, the contractor shall document a detailed plan of the procedures to be employed during handling of the munition. This documentation shall describe, as a minimum, the procedures to be employed to assure that:

1. The shorting bar is not deformed or broken.

2. The battery is not activated.

3. The two munition models remain segregated at all times.

4. The exterior model markings on each mine are consistent with the munitions it contains.

5. The temperature of the molded assemblies, or the munition does-not exceed 165°F at any time during processing or storage.

6. The temporary safety pin is present.

7. The mounting plate insulator is properly installed.

8. The special test mines shall always be marked and segregated from the other munition models.

The plan shall be submitted to the address specified in 6.7 for approval. Approval shall be received prior to any loading. Any changes to the approved plan shall be submitted for approval prior to implementation.

3.4.1 Documentation of controls for use of adhesives and sealants. The contractor shall develop a documented operational procedure for utilization of adhesives and sealants. This procedure shall provide for control of the materials and processes to be used; including controls for the following: shelf life, handling, mixing, curing and pot life. Necessary cleaning agents, safety precautions, ventilation, and any other information considered necessary for the proper use of the adhesive(s) or sealant(s) shall also be included. The procedure shall be updated whenever a change is made to the process such that product quality may be affected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:59Z
Check the source to verify that this is the current version before use.

3.4.2 Documentation of plan for molding and cure process controls. The contractor shall prepare a written plan to be implemented for control of the process for molding the kill mechanism and overlay assembly into the housing, timing, and fuzing. The plan shall breifly describe the entire process and shall fully document the controls to be implemented for maintaining the process. The controls shall include, as a minimum, all the checkpoints referenced in 6.4. Submission of this plan shall be in accordance with Data Item Description.

Approval of the plan shall be limited to the controls to be employed and not the overall potting process, nor shall such approval relieve the contractor, in any way, of the responsibility for producing acceptable product.

3.5 Compressive strength of molding compound. The molding compound for assembling the plug to the housing shall comply with MIL-M-48245. The resin/hardener weight ratio shall be 2.0/1.0 ± 5% (1.8/1 to 2.2/1) and the ultimate compressive strength shall be 29,000 to 38,200 psi.

3.6 Temperature control during molding and cure of munition. The munition shall not be subjected to temperatures exceeding 165°F at any time.

3.7 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.8 Workmanship. All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust and other foreign matter that may impair proper functioning. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agents. All required marking and stamping shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may utilize his own or any other facilities sutiable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:59Z
Check the source to verify that this is the current version before use.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual nor does it commit the Government to acceptance of any defective material.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item.

a. First Article Inspection (see 4.3)

b. Quality Conformance Inspection (see 4.4)

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of-the following items in sample quantities as indicated.

Part description	Drawing	Quantity
Special Test Mine	9366705	12
Test bars		2
Bag Assembly	12910118	12

4.3.2 inspections to be performed. See MIL-A-48076 and Table I herein.

4.3.3 Rejection. See MIL-A-48078. The first article sample shall be rejected if any mine fails to comply with any of the examinations or tests specified in this specification or any of the requirements of the applicable drawings.

In addition, the first article sample shall be rejected if the mines any one of the following defects:

Any premature arming of the S&A firing of the gas generator or the piston actuator.

b. The shorting bar is cracked or broken prior to arming.

c. The safety pin is missing (prior to testing).

d. The safety clip is missing (prior to testing).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:59Z
Check the source to verify that this is the current version before use.

TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-70765 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Mine				See below
					NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Special Test Mine (Dwg. 9366705)				
	Examination for defects	10		3.2	4.4.2.7
	Compressive Strength of Test bars			3.6	4.5.1
	Functioning Test	72		3.3	4.5.1.3

NOTES:

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:59Z
Check the source to verify that this is the current version before use.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of projectiles shall contain:

a. Arming assembly metal parts assemblies of one lot interfix number from one manufacturer.

b. Molded assembly of one lot serial number form one manufacturer and of one designated type.

4.4.2 Examination. See MIL-A-48078.

4.4.2.1 Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accorance with MIL-STD-105, Inspection Level II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:59Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-70765 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Arming Assembly, PDM	1 OF 1	12579595
			NEXT HIGHER ASSEMBLY: 12579601

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Presence of all parts		0.40%	3.2	Visual
102	Coated ring properly seated in arming lever		0.40%	3.2	Visual
Minor					
201	Poor workmanship		1.0%	3.8	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:59Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-70765 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Pin, Safety, Temporary				12579592
					NEXT HIGHER ASSEMBLY 12579596
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
1	Material per print		each lot	3.1	Lab verification
2	Length of major diameter		100%		Gage
3	Diameter of flat section		100%		Gage
Major	None defined				
Minor					
201	Poor workmanship		1.0%	3.8	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:59Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-70765 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Plate, Safety, Temporary				12579593
					NEXT HIGHER ASSEMBLY 12579596
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Material per print		each lot	3.1	Certification
Minor 201	Poor workmanship		1.0%	3.8	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:59Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-70765 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Kill Mechanism Assembly, Inert				9298442
					NEXT HIGHER ASSEMBLY 9298449
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Outside diameter, max.		0.65%	3.2	Gage
202	Total height, max.		0.65%	3.2	Gage
203	Poor workmanship		1.0%	3.8	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-02-23T03:59Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-70765 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.5	Kill Mechanism and Overlay Assembly (Inert)		9298449
			NEXT HIGHER ASSEMBLY: ---

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1	Presence of any explosive component		100%	3.2	Visual
Major 101	Diameter, max (at equator)		0.40%	3.2	Gage
Minor 201	Poor workmanship		1.0%	3.8	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:59Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-70765 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Molded Assembly (Prior to assembly of inert kill mechanism and overlay assembly)				9366705
					NEXT HIGHER ASSEMBLY ---
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
a.	Temporary safety pin not present or not properly positioned in S&A		100%	3.2	X-ray
b.	Shorting bar broken or damaged				
Major					
101	Battery ball depressed below normal position		100%	3.2	Gage
102	Foreign material in kill mechanism cavity		0.40%	3.2	Visual
103	Largest diameter in kill mechanism cavity not prepared for bonding		0.40%	3.2	Visual
104	Sensor plug(s) missing or damaged		0.40%	3.2	Visual
Minor					
201	Poor workmanship		1.0%	3.8	Visual

NOTES:

(((

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:59Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-70765 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.7	Special Test Mine				9366705
					NEXT HIGHER ASSEMBLY ---
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Shorting bar broken or damaged		100%	3.2	Visual
2	Safety pin properly installed		100%	3.2	Visual
3	Safety clip properly installed		100%	3.2	Visual
4	Hazardous dud (LAT)		100%	3.3.3	4.5.3.1
5	Premature arming		100%	3.3.3	4.5.3.1
6	Lead balls missing in KM		100%	3.3.1	Gage
Special					
1	Mounting plate insulator properly installed		100%	3.2	Visual
2	Groove missing or arc face		100%	3.2	Visual
3	No space or interfrume fit between cam and battery ball		100%	3.2	Gage
Major					
101	Arming mechanism functioning (LAT)			3.3.1	4.5.3.1
102	Anti-disturbance functioning (LAT)			3.3.4	4.4.3.1
103	Breakwire functioning (LAT)			3.3.5	4.4.3.1
104	Tripline deployment (LAT)			3.3.8	4.4.3.1
105	Safe duds (LAT)			3.3.2	4.5.3.1
106	Early functions (LAT)			3.3.7	4.4.3.1
107	Self-destruct functioning (LAT)			3.3.63	4.4.3.1
108	Distance between battery ball and cam		0.40%	3.2	Std. gage

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-02-23T03:59Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-70765 (AR)

PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER
4.4.2.7	Special Test Mine		9366705
			NEXT HIGHER ASSEMBLY: ---

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major	(continued)				
109	Any sensor restricted (cap not removed or molding compound/flash over sensor cutout)		0.40%	3.2	Visual
110	Any sensor prematurely deployed		0.40%	3.2	Visual
111	Mounting screws properly torqued		0.40%	3.2	Torque wrench
112	Shorting bar hook attached per print		0.40%	3.2	Visual
113	Shorting bar clip attached per print		0.40%	3.2	Visual
114	Nose protector missing		0.40%	3.2	Visual
115	Tape properly intalled		0.40%	3.2	Visual
Minor					
201	Test pads exposed		1.0%	3.2	Visual
202	Poor workmanship		1.0%	3.8	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:59Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-70765 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Packing and Marking for Box, Wirebound for Mine, APERS, M86 Classroom Model, Special Test Mine or Demonstration Mine	1 OF 1	9366710
			NEXT HIGHER ASSEMBLY 9366711

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Packing material missing		0.40%	3.2	Visual
102	Incorrect packing material used		0.40%	3.2	Visual
103	Packing material installed incorrectly		0.40%	3.2	Visual
104	Marking incorrect or missing		0.40%	3.2	Visual
105	Wirebound box closed incorrectly		0.40%	3.2	Visual
Minor					
201	Poor workmanship		1.0%	3.8	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:59Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.9	Packing and Marking for Shipping and Storage Container, PA19 for Mine, APERS, M86 Classroom Model, Special Test Mine or Demonstration Mine	1	1	9366709
				NEXT HIGHER ASSEMBLY: 9366710

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
1	Marking on PA19 container incorrect		100%	3.2	Visual
Major					
101	Packing material(s) incorrect sizes		0.40%	3.2	Visual
102	Moisture in packing material above max.		0.40%	3.2	Visual
103	Outer barrier bag marked incorrectly		0.40%	3.2	Visual
104	Mine inserted incorrectly into inner bag and bandoleer		0.40%	3.2	Visual
105	Outer bag loaded incorrectly		0.40%	3.2	Visual
106	Outer bag sealed incorrectly		0.40%	3.2	Visual
107	Pull tabs missing from outer bag		0.40%	3.2	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-02-23T03:59Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-70765 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.10	Box, Wirebound		9366708
			NEXT HIGHER ASSEMBLY 9366710

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Incorrect material		0.40%	3.2	Visual
102	Inside tolerances incorrect		0.40%	3.2	Visual
Minor					
201	Poor workmanship		0.40%	3.8	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:59Z
Check the source to verify that this is the current version before use.

4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection Tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examination and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit inspection equipment designs for approval in accordance with the terms of the contract. See Section 6 of MIL-A-48078.

4.5 Methods of inspection.

4.5.1 Quality controls for each molding compound batch for assembling plugs to housings.

4.5.1.1 Molding compound batch definition. A batch of molding compound is herein defined as that quantity of compound mixed at one time using base resin and hardener from one lot interfix number from one manufacturer and used to assemble plugs to housings in one unchanged process. When a continuous type mixing process is used, a batch is defined as that quantity of molding compound mixed and dispensed in one continuous unchanged process during a four (4) hour period or portion thereof using base resin and hardener from one lot interfix number from one manufacturer.

4.5.1.2 Resin/hardener weight ration control.

a. Hand mixing: For each batch of molding compound mixed at one time, the hardener and resin shall be weighed separately to verify that the ratio is between 1.8/1 and 2.2/1 before use.

b. Automatic mixer method: At the beginning of production and one each four (4) hours during production? two test shots shall be drawn with the mixer set to expel the hardener and resin separately. The hardener and resin shall be weighed to verify that the ratio is between 1.8/1 and 2.2/1. If the shots fail to meet the requirement, all wedges filled since the last acceptable test shall be rejected. Further, the mixer shall be corrected and shall pass the above test before being returned to use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:59Z
Check the source to verify that this is the current version before use.

4.5.1.3 Cure control. Two test bars (1/2 inch x 1/2 inch x 1.6 inches) from each batch of molding compound shall be molded at the same time the batch is used to assemble plugs to housings. Test bar molding shall be by best commercial practice, including vibration and evacuation if necessary. The mix weight ratio limits shall be 2/1 ± 5% (1.8/1 to 2.2/1). The test bars shall be processed with the items assembled from the batch of molding compound which the bars represent. , lie. the bars shall be cured in the same ovens and subjected to the same temperatures for the same time periods as the items they represent. After completion of the cure, the bar shall be tested for ultimate compressive strength as specified in ASTM-D-695 except that the test bar dimension shall be 0.5 inches by 0.5 inches by 1.6 inches. The ends shall be flat and parallel to within . 001 inch and shall be perpendicular to the long axis of the test bar. Speed of testing shall be as specified in the ASTM. If the results of this test are outside the limits specified in 3.6, the munitions represented by the bar shall be rejected.

4.5.1.4 Heat seal test. The contractor shall develop a heat seal test to detect any air leakage into the sealed bag assemblies . As a minimum the test shall be in accordance with the applicable test specified in MIL-P-116. Any test plan used shall be submitted to: ATTN: AMSMC-QAT-M (D) , Picatinny Arsenal, NJ 07806-5000 for approval.

4.5.2 Temperature control during molding/cure operations. Whenever the housings *or* mine is subjected to heating in ovens, temperature records shall be maintained and reviewed to assure that the minimum temperature has been reached and that the maximum temperature has not been exceeded. If any units have been subjected to temperatures beyond the maximum limit at any time, regardless of duration, they shall be rejected. Temperature measuring instruments shall be approved and calibrated in accordance with 4.4.4.

In addition, if units are stored where temperatures may exceed the limit specified in 3.7, temperature monitoring equipment shall be provided and constant records maintained of the temperature to assure that the temperature limit is not exceeded.

4.5.3 Functioning.

4.5.3.1 Testing. This test shall be conducted at a Government proving ground or a place determined by the contract. The mine shall be deployed by a remote arming fixture. The forces necessary to pull the safety clip and aim the mine will be recorded.

After deployment and impact, a record shall be made of the number of sensors deployed from each munition and approximate length of each tripline. Then one-third of the mines shall be tested in the breakwire mode, one-third shall be tested in the anti-disturbance (AD) mode, and one-third shall be tested in the self-destruct mode. Any unit which fails to deploy any sensors shall be tested in the self destruct mode.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:59Z
Check the source to verify that this is the current version before use.

Those units to be tested in the breakwire mode shall be tested by restricting movement of the item and attaching a force gage to a tripline and exerting an axial pull force to function the item. A record shall be made of the force required to function the breakwire. Units which fail to function on first attempt shall be tested in the AD mode.

After depressing the battery ball initiation of the arming cycle) monitoring of the mine is required to insure that the mine did not fire the detonator, piston actuator or gas generator prior to 25 seconds after depressing of the battery ball.

Those units to be tested in the anti-disturbance mode shall be tested by rolling the item from one face to another. Units which fail to function on the first attempt shall be tested in the breakwire mode.

Recording equipment may be used to determine the time of functioning of items tested in the self-destruct time. After the maximum time-out time has elapsed plus one hour, the impact area shall be searched for non-functioning munitions. Any munition which has not functioned shall be treated as a hazardous dud until it can be determined that the item is not capable of functioning. This shall be determined as follows using suitable protection and safety precautions for personnel handling the munitions:

perform visual observation to assure the battery ball is depressed, the shorting bar is sheared and to determine if there is any evidence that the detonator has functioned.

b. If it cannot be determined whether or not the detonator has functioned, attempts shall be made to function the item by tipping about an edge, pulling all deployed triplines.

If the detonator functions at any time, the item shall be classed as a hazardous dud.

c. If the item has not functioned after the above external inputs have been applied, it shall be placed in an approved explosive handling box and removed from the impact area.

d. After a minimum sterilization period from time of impact, the item shall be remotely removed from the box and examined. If there is any evidence that the detonator has functioned, the item shall be classed as a hazardous dud. If the item has not functioned it shall be examined to determine if the battery ball has been depressed. Normally, the ball is depressed approximately flush with the apex radius. If the ball is depressed (and the shorting bar sheared) and there is not evidence of functioning, the item shall be dropped as specified in b. If the item still has not functioned, it shall be classed as a non-hazardous dud.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:59Z
Check the source to verify that this is the current version before use.

If it can be determined that the battery has not been activated, the item shall be classed as a hazardous dud unless it can also be determined that the S&A has not armed, the safety pin can bve fully inserted back.

4.5.3.2 Records of tests. A record shall be made for each mine tested and shall include an accounting for the defects listed in 4.4.2.24 or below as applicable. In addition, the record shall include malfunctions, actual munition functioning times including early functions? sensor deployment of each munition, anti-disturbance and breakwire functioning performance units which fail to function on first attempt, etc, and any other pertinent data. A copy of the test results for each lot shall be forwarded to the address specified in 6.6. The lot shall be rejected if any one of the following occurs:

a. Any premature firing burst occurs (see 4.5.7.1).

b. Any shorting bar is not completely removed or sheared or any battery ball is not completely depressed.

c. Any munition functions (or is capable of functioning) after the maximum time has elapsed (Hazardous dud).

d. Muntions do not meet required functioning rate.

5. PACKAGING

5.1 Unit Packaging.

5.1.1 Level A. Level A unit packaging shall be in accordance with Dwg. 9366709.

5.1.2 Level B. Level B unit packaging Shall have the same requirements as for Level A.

5.1.3 Level C. Level C unit packaging shall have the same requirements as for Level A.

5.2 Packing.

5.2.1 Level A. Level A packing shall be in accordance with Dwg. 9366710.

5.2.2 Level B. Level B packing shall have the same requirements as for Level A.

5.2.3 Level C. Level C packing shall have the same requirements as for Level A.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:59Z
Check the source to verify that this is the current version before use.

5.3 Marking.

5.3.1 Level A. Level A marking shall be in accordance with Dwg. 9366710 and Dwg. 9355710.

5.3.2. Level B. Level B marking shall have the same requirements as for Level A.

5.3.3 Level C. Level C marking shall have the same requirements as for Level A.

5.4 Shipping. When shipments of units from more than one lot are made at one time, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use. This specification is intended for use in loading assembling and packing the Mine, Anti-Personnel, M86, Special Test Mine.

6.2 ordering data. See MIL-A-48078.

6.3 Submission of designs for approval. See MIL-A-48078. Submit equipment designs as required to Commander, Picatinny Arsenal, ATTN: AMSMC-QAT-M (D) , Picatinny Arsenal, NJ 07806-5000. This address iwll be specified on the Contract Data Requirements List, DD Form 1423 in the Contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Documentation of molding process controls. The contractors's documentation of molding process shall specify, as a minimum, controls and procedures for the following:

a. Confirmation of accuracy of temperature-measuring instruments, including the method, location and. frequency basis for obtaining temperature checks of all the temperature recording instruments for all the molding and curing phases.

b. Homogeneity of molded munitions, to preclude mixing of munitions from different lots during molding and cure.

c. The storage of permanent records, available for Government review, of the time-temperature cycle for all phases of the molding and cure process and for the items in a and b above. The records shall provide suitable identification by molding compound batch and lot number as applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:59Z
Check the source to verify that this is the current version before use.

6.5 Safety precautions for molding compound and sealing compound. When handling, mixing or curing the molding or sealing compound or its constituents, suitable precautions must be taken, as some of the constituents and compounds are toxic and exude toxic fumes when mixed or heated. Use adequate ventilation. Avoid contact with skin, eyes and clothing and prolonged repeated breathing of fumes.

6.6 Data cards. In addition to the normal distribution of records, on copy of the ammunition data card for each munition lot shall be forwarded to: Commander, US ARDEC, ATTN: AMSMC-QAT-M (D), Picatinny Arsenal, New Jersey 07806-5000.

6.7 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center, (ARDEC), may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, of Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.8 Data requirements. Deliverable data required by this specification is cited in the paragraphs listed below. The Form DD 1423 should include distribution to AMSMC-QAT-M (D).

Paragraph	Data Requirements	Applicable DID
3.41	Plan for controls in use of Adhesives and sealants	DI-T-5372 (Tailored)
3.4	Plan for Safety in handling and Processing of Munitions	DI-T-5372 (Tailored)
3.42	Plan for Molding and Cure Process Controls	DI-T-5372 (Tailored)
4.4.2	Quality Inspection Defect Report	DI-R-1721
6.6	Ammunition Data Card	DI-L-1410

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:59Z
Check the source to verify that this is the current version before use.

6.9 Submission of results of contractor-conducted examinations and tests. Data shall be submitted in accordance with data item DI-R-1721 on the DD Form 1423 for the contract.

6.10 Mine functioning.

6.10.1 Premature. A premature functioning is considered to have occurred if either the detonator, piston actuator or gas generator functions in less than 25 seconds.

6.5.2 Functioning. Satisfactory functioning of the mine is defined as successful arming and explosive functioning of the S&A after 25 seconds.

6.6 Submission of test data. In addition to the normal distribution of records, when the mines are procured by the US AMCCOM, one (1) copy of all data and ammunition data cards shall be forwarded to: Commander, Picatinny Arsenal, ATTN: AMSMC-QAT-M (D), Picatinny Arsenal, NJ 07806-5000.

6.7 Submission of plans for approval. The plans required in Section 3 shall be submitted to the following address for approval: Commander, Picatinny Arsenal, ATTN: AMSMC-QAT-M (D), Picatinny Arsenal, NJ 07806-5000.

6.8 Safety precautions for molding compound and bedding compound. When handling, mixing or curing the molding or bedding compound or its constituents and compounds are toxic and exude toxic fumes when mixed or heated. Use adequate ventilation. Avoid contact with skin, eyes and clothing and prolonged or repeated breating of fumes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1320-A512)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:59Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-70765	MINE ANTI-PERSONNEL, M86 SPECIAL TEST MINE LOADING AND ASSEMBLY F

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:59Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Armament Munitions &
Chemical Command
ATTN: AMSMC-QA
Picatinny Arsenal, NJ 07806-5000

Source: https://assist.dla.mil -- Downloaded: 202610-23T03:59Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:59Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:59Z
Check the source to verify that this is the current version before use.

NOTICE OF INACTIVATION
FOR NEW DESIGN

INCH-POUND

MIL-M-70765(AR)
NOTICE 1
7 February 1996

MILITARY SPECIFICATION

MINE, ANTI –PERSONNEL M86: SPECIAL TEST MINE LOADING AND ASSEMBLY OF

This notice should be filed in front of MIL-M-70765(AR), dated 10 July 1988

MIL-M-70765[AR). dated 10 July 1988,is inactive for new design and is no longer used, except for replacement purposes.

Preparing Activity:
Army – AR

AMSC N/A FSC 1345

DISTRIBUTION STATEMENT A Approved for public release: distribution is unlimited

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:59Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:59Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:59Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:59Z
Check the source to verify that this is the current version before use.